Ansible Tasks -03

- Write a single ansible playbook which will install apache and nginx. Note: Playbook should not be hardcoded and pass the variables from a different file.

```
[ec2-user@ip-172-31-24-112 playbooks]$ cat services.yml
- hosts: all
  become: yes

  vars_files:
    - variables.yml

  tasks:
    - name: Install packages
      dnf:
        name: "{{ item }}"
        state: present
      loop: "{{ packages }}"
```

```
[ec2-user@ip-172-31-24-112 playbooks]$ cat variables.yml
files:
  - Finally.txt
  - done_with.txt
  - the_task.txt
packages:
  - httpd
  - nginx
```

```
[ec2-user@ip-172-31-24-112 playbooks]$ ansible-playbook services.yml

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [web1]
ok: [web2]

TASK [Install packages] ********************************************************************************************
ok: [web2] => (item=httpd)
ok: [web1] => (item=httpd)
changed: [web1] => (item=nginx)
changed: [web2] => (item=nginx)

PLAY RECAP *********************************************************************************************************
web1                       : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web2                       : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
[ec2-user@ip-172-31-29-27 ~]$ systemctl status nginx
o nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; p
     Active: inactive (dead)
```

```
[ec2-user@ip-172-31-31-155 ~]$ sudo systemctl status nginx
o nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabl
     Active: inactive (dead)
```

- Ansible playbook to create 10 different directories with minimal code, and directory names should be passed as variables.

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```
[ec2-user@ip-172-31-24-112 playbooks]$ cat dir.yml
- hosts: all
  become: yes

  vars_files:
    - variables.yml

  tasks:
    - name: Create directories
      file:
        path: "/tmp/{{ item }}"
        state: directory
      loop: "{{ dirs }}"
```

```
[ec2-user@ip-172-31-24-112 playbooks]$ ansible-playbook dir.yml

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [web2]
ok: [web1]

TASK [Create directories] ******************************************************************************
changed: [web1] => (item=dir1)
changed: [web2] => (item=dir1)
changed: [web1] => (item=dir2)
changed: [web2] => (item=dir2)
changed: [web2] => (item=dir3)
changed: [web1] => (item=dir3)
changed: [web2] => (item=dir4)
changed: [web1] => (item=dir4)
changed: [web1] => (item=dir5)
changed: [web2] => (item=dir5)
changed: [web2] => (item=dir6)
changed: [web1] => (item=dir6)
changed: [web2] => (item=dir7)
changed: [web1] => (item=dir7)
changed: [web2] => (item=dir8)
changed: [web1] => (item=dir8)
changed: [web2] => (item=dir9)
changed: [web1] => (item=dir9)
changed: [web2] => (item=dir10)
changed: [web1] => (item=dir10)

PLAY RECAP *********************************************************************************************
web1                       : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web2                       : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
[ec2-user@ip-172-31-29-27 tmp]$ ls
dir1   dir2  dir4  dir6  dir8  syste
dir10  dir3  dir5  dir7  dir9  syste
```

```
[ec2-user@ip-172-31-31-155 tmp]$ ls
dir1   dir2  dir4  dir6  dir8  syste
dir10  dir3  dir5  dir7  dir9  syste
```

- Ansible playbook to copy ssh-keygen from master to worker nodes. Note:

  a) Provision new 3 ec2 machines, one master and two worker nodes.

  b) Create common user called ansadm and provide sudo priviliges on 3 ec2 instances.

  c) Create ssh-keygen in master and your playbook should copy the keygen making it password less authentication.

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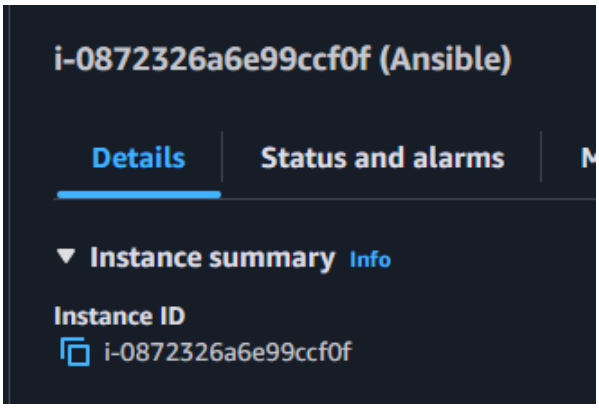

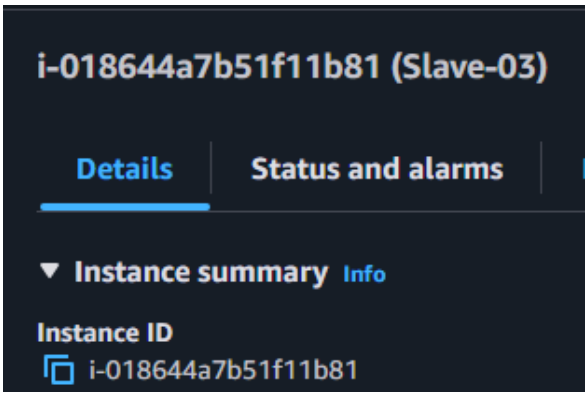

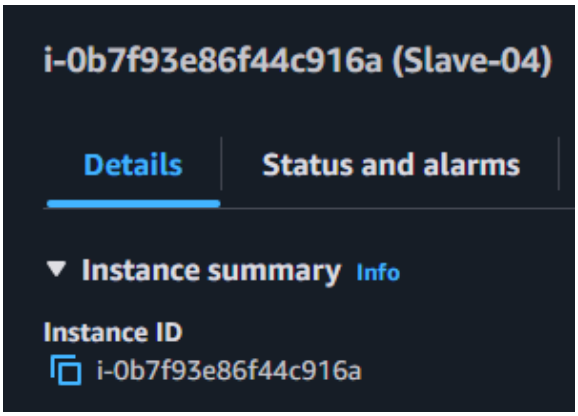


```
[ec2-user@ip-172-31-24-112 playbooks]$ cat ssh.yml
- hosts: web3,web4
  become: yes

  tasks:
    - name: Copy controller public key
      authorized_key:
        user: ec2-user
        state: present
        key: "{{ lookup('file', '~/.ssh/id_ed25519.pub') }}"
```

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```
[ec2-user@ip-172-31-24-112 playbooks]$ ansible-playbook ssh.yml --private-key ~/Linux-key.pem

PLAY [web3,web4] ***************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [web3]
ok: [web4]

TASK [Copy controller public key] **********************************************************************************
changed: [web3]
changed: [web4]

PLAY RECAP *********************************************************************************************************
web3                       : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web4                       : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
[ec2-user@ip-172-31-24-112 playbooks]$ cat user.yml
---
# This playbook runs on all servers in the inventory
- hosts: all

  # become: yes means tasks run with sudo privileges
  become: yes

  vars_files:
    - variables.yml

  tasks:

    # Task 1: Create a user called ansadm
    - name: Create ansadm user
      user:
        name: "{{ username }}"        # username to create
        state: present      # ensures the user exists
        groups: wheel       # wheel group gives sudo access in Amazon Linux 2023
        create_home: yes    # creates /home/ansadm directory

    # Task 2: Give the user passwordless sudo access
    - name: Allow ansadm sudo access
      lineinfile:
        path: /etc/sudoers                    # sudo configuration file
        line: "{{ username }} ALL=(ALL) NOPASSWD:ALL" # allows ansadm to run sudo without password
        validate: visudo -cf %s               # checks syntax before saving to avoid breaking sudo
```

```
[ec2-user@ip-172-31-24-112 playbooks]$ ansible-playbook user.yml

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [web3]
ok: [web2]
ok: [web1]
ok: [web4]

TASK [Create ansadm user] ******************************************************************************************
changed: [web3]
changed: [web4]
changed: [web2]
changed: [web1]

TASK [Allow ansadm sudo access] ************************************************************************************
changed: [web2]
changed: [web3]
changed: [web4]
changed: [web1]

PLAY RECAP *********************************************************************************************************
web1                       : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web2                       : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web3                       : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web4                       : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

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```
## Read drop-in files from /etc/sudoers.
#includedir /etc/sudoers.d
ansadm ALL=(ALL) NOPASSWD:ALL
[ec2-user@ip-172-31-29-27 ~]$
```

```
## Read drop-in files from /etc/sudoer
#includedir /etc/sudoers.d
ansadm ALL=(ALL) NOPASSWD:ALL
[ec2-user@ip-172-31-31-155 ~]$
```

```
## Read drop-in files from /etc/sudoe
#includedir /etc/sudoers.d
ansadm ALL=(ALL) NOPASSWD:ALL
[ec2-user@ip-172-31-18-55 ~]$
```

```
## Read drop-in files from /etc/sudoe
#includedir /etc/sudoers.d
ansadm ALL=(ALL) NOPASSWD:ALL
[ec2-user@ip-172-31-31-150 ~]$
```

- Ansible playbook to inject ansible vault variables.

```
[ec2-user@ip-172-31-24-112 playbooks]$ cat secret.yml
- hosts: all
  become: yes

  vars_files:
    - secrets.yml

  tasks:
    - name: Use the secret
      debug:
        msg: "{{ db_password }}"
```

```
[ec2-user@ip-172-31-24-112 playbooks]$ ansible-playbook secret.yml --ask-vault-pass
Vault password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [web2]
ok: [web4]
ok: [web3]
ok: [web1]

TASK [Use the secret] **********************************************************************************************
ok: [web1] => {
    "msg": "starktech123"
}
ok: [web2] => {
    "msg": "starktech123"
}
ok: [web3] => {
    "msg": "starktech123"
}
ok: [web4] => {
    "msg": "starktech123"
}

PLAY RECAP *********************************************************************************************************
web1                       : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web2                       : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web3                       : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web4                       : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```